A Novel Motor Imagery Decoding Method Based on MODWT and Light-weight CNN

Wei Duan
Institute of Software, Chinese Academy of Sciences
Beijing, China
duanwei@iscas.ac.cn

Tinglei Huang*
Institute of Software, Chinese Academy of Sciences
Beijing, China
Guangxi Academy of Sciences
Nanning, China
tinglei@iscas.ac.cn

Jie Liu
Institute of Software, Chinese Academy of Sciences
Beijing, China
liujie2016@iscas.ac.cn

2024 IEEE First International Conference on Data Intelligence and Innovative Application (DIIA) | 979-8-3315-0443-4/24/$31.00 ©2024 IEEE | DOI: 10.1109/DIIA62678.2024.10871526

***Abstract*—Motor imagery (MI) is one of the most important brain computer interface (BCI) paradigms, which has been widely used in many clinical and scientific applications, such as rehabilitation and motion control. Despite the fast development of deep learning method, MI electroencephalography (EEG) signal decoding has remained a critical challenge to because of its poor SNR, low spatial resolution, large inherent variability, and small dataset size. In this work, we propose a maximal overlap discrete wavelet transform (MODWT) and light-weight convolution neural network (CNN) combined method, which effectively improves EEG decoding performance with relatively fewer network parameters by incorporating time-frequence and time-spatial features of EEG. Besides, the method employs channel attention mechanism to enhance important features and reduce redundance. We conduct the experiment on the public dataset BCI Competition IV 2a, and the results show that the proposed method can achieve better performance than other light-weight model, with 78.7% accuracy (Kappa 0.71) with four-class classification, which well validates the effectiveness and robust of the proposed method.**

Keywords—BCI, Motor imagery, EEG, Decoding, CNN, MODWT

I. INTRODUCTION

With the fast development of neuroscience, brain computer interface (BCI) has been the most promising frontier technology recently, which revolutionizes the interaction between human brain and external environment. Motor imagery(MI) is the mental rehearsal of a physical movement without actual execution [1], and has become one of the most important BCI paradigms due to its ability to offer a non-invasive and more intuitive way for human to control external devices. Electroencephalography (EEG) provides an convenient way for recording neural activity with low cost and high temporal resolution [2], and it has attracted much attention in MI-based BCI systems. Through decoding MI EEG signal, MI-based BCI allows external device being controlled by human's thought. MI with EEG has shown broad prospect in motor rehabilitation, assisting the paralyzed and smart control [3].

While the key to the success of MI-based BCI application lies in accurately decoding and classification of EEG signals, it has posed a great challenge to existing methods, which is primarily caused by the vulnerability to environment, low SNR, and individual differences. Traditional decoding methods are based on the machine learning algorithms. First, the EEG signal is projected on the feature space by certain transformation, such as independent component analysis (ICA), power spectral density (PSD), etc. And then the linear discriminant analysis (LDA) or support vector machines (SVM) is used as classifier to sperate different groups. Among those, common spatial patterns with filter bank (FBCSP) has shown as the most effective classical method in MI decoding [4], which is based on the differences between multiple frequency bands. However, the performance of these traditional decoding methods are strongly limited by the low spatial resolution, cross-subject variability and the small dataset.

With the fast development of deep learning (DL) techniques, DL has been demonstrated highly effective in computer vision and natural language processing, and it has greatly inspired MI EEG decoding with high accuracy and performance. A variety of novel MI decoding methods based convolution neural network (CNN) are proposed in recently years, and they have shown significant advantages on automatically features extracting and sophisticated pattern recognition over traditional methods. The Deep and Shallow ConvNet proposed by Schirrmeister [5] leveraged two or more convolution layers with large kernels to extract temporal and spatial features for EEG signal. Qin et al. employed the efficient channel attention (ECA) and temporal convolutional network to enhance the extraction of channel-specific features [6]. Cai et al. have introduced MT-MBCNN [7] based on the combination of multibranch spatial-temporal feature and multi-bands spectral feature. These methods have made remarkable achievements on EEG decoding, but they require larger model to obtain high classification accuracy, resulted in the problem of long training time and overfitting. Many BCI applications have high requirements on real-time, stability and reliability, and some light-weight networks have also shown excellent performance on EEG decoding. Lawhern have proposed a compact CNN network, EEGNet [8], which used the depthwise convolution and separable convolution to efficiently extract temporal-spatial features. Mane et al. introduced FBCNet [9], a filter bank CNN model, and it effectively capture and utilize the rich frequency of the multiple frequency sub-bands. However, current methods didn't take full advantage of the inherent temporal, spatial and spectral features of EEG signal.

Motivated by previous research, we propose a novel MI decoding method based on MODWT and light-weight CNN network (MODWT-Net) in this work. It uses the channel attention mechanism to join the original EEG signal and its time-frequency signal together. Besides, this network incorporates the time-frequency and temporal-spatial features through multiple temporal convolution layer. The MODWT-Net is validated on the BCI Competition IV 2a dataset, and

This research was funded by the STI 2030—Major Projects 2021ZD0200403, and STI 2030—Major Projects 2021ZD0200800.

the classification results have demonstrated the effectiveness of the proposed method.

II. METHODS

We develop the MI-EEG classification method by integrating MODWT and light-weight CNN model, and the architecture is shown as Figure 1. The MODWT-Net consists of three parts: (1) time-frequency analysis block, (2) temporal convolution block, (3) spatial convolution block. The first block is employed to extract time-frequency features of EEG signal, and fuse the temporal features and time-frequency features with attention mechanism. The temporal convolution block is designed to obtain the multi-scale temporal features through several convolution layers. And the spatial convolution block is to extract spatial information from different EEG channels. Finally, the classifier completes the EEG classification by the flattened fusion features.

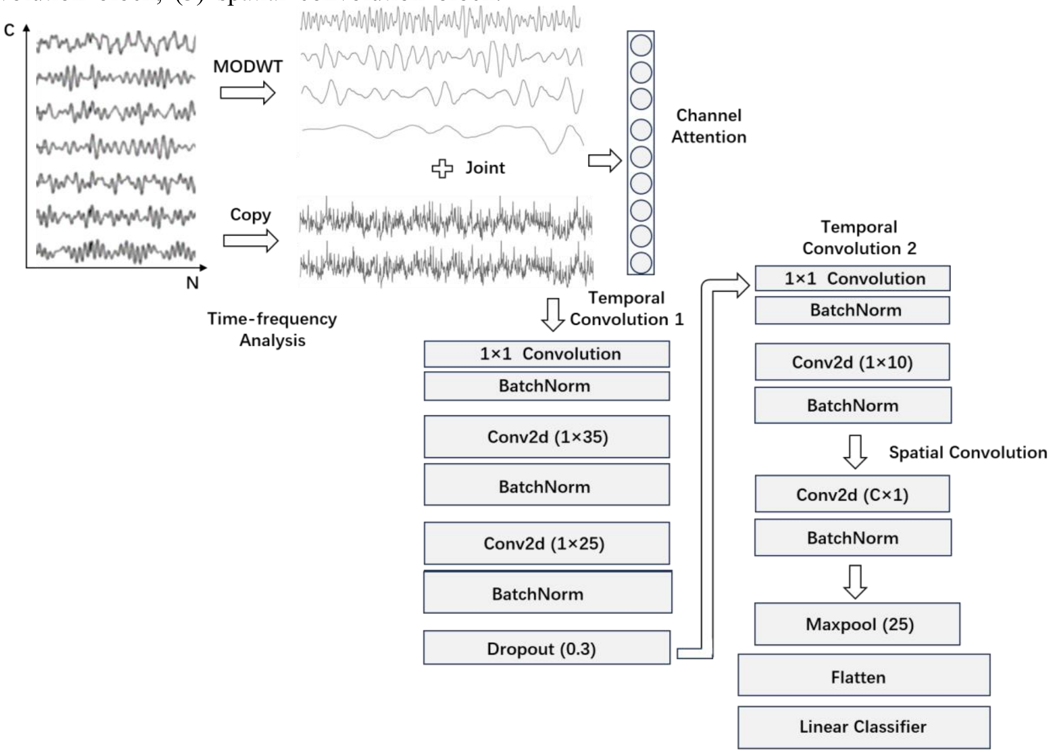

Fig. 1. The architecture of the proposed method.

A. Time-frequency Analysis Block

MI EEG signal is highly non-stationary and non-linear, and it cannot be adequately expressed by temporal or spectral features separately. The time-frequency analysis can provide insights into the temporal dynamics of EEG signal [10]. MODWT is a type of wavelet transform that retains the properties of time invariance. Unlike discrete wavelet transform (DWT), MODWT employs the maximize overlapping method in the wavelet and scaling function to decompose the signal into different layers without down sampling the original signal. Thus, it can obtain the time-localized frequency information, and capture both low-frequency trends and high-frequency fluctuations, which is more suitable for extracting the local detail of EEG signal. MODWT can be calculated by follows:

$$w_j(k) = \sum_{n=0}^{L_j-1} x(n)h(n-2^j k) \quad (1)$$

$$v_j(k) = \sum_{n=0}^{L_j-1} x(n)g(n-2^j k)$$

where $x(n)$ is the original signal, $g(n)$ and $h(n)$ are the scaling and wavelet function, $L_j$is the length of j-th signal, w_j and v_j are the j-th wavelet decomposition coefficient. In this work, we use the sysm4 wavelet function and decompose the signal into five layers.

As EEG signal has strong inter-subject variability, some MI EEG signal has shown less significant on spectral domain. To improve model generalization, we combine the decomposed signal and the original EEG signal, after MODWT decomposing the preprocessed EEG signal. And the dimension of the input signal has changed from (1, C, T) to (D, C, T), where D is the joint layer number of decomposed layers and original signal layers. Besides, the different layers of decomposed and original signals should be put on different weight. Thus, the channel attention mechanism is utilized to dynamically focus on the most subject-related features. The input signal is defined as X, W is the weighted function, and the channel attention mechanism will be given as follows:

$$X^{new} = X_{D,C,T} \odot W_D \quad (2)$$

The time-frequency analysis block conducts the multi-resolution analysis for MI EEG signal, and then effectively integrates the time-frequency features and subsequent temporal-spatial features, which will help the network better to learn the most important features of EEG signal.

B. Temporal Convolution Block

EEG signal has rich temporal and spatial information, and the temporal convolution block plays an key role in MI EEG classification tasks. Multiple temporal convolution layers are used in this block to capture the dynamic features of EEG, which can enrich the diversity of high-dimensional features. Moreover, to reduce the complexity of subsequent spatial convolution network, EEG signal should be processed through temporal convolution with large kernel before fed into the spatial convolution block. In the temporal convolution block 1, we firstly utilize 1×1 convolution to increase the dimension of the input EEG and decomposed joint signal. Then two cascaded conv_2d layers with different kernel size are employed to extract temporal features, followed by the Dropout layer to avoid the overfitting. Besides, the depthwise convolution and pointwise convolution are used to reduce the trainable network parameters and computations. In the temporal convolution block 2, 1×1 convolution is used to reduce the data dimension while maintaining the feature map. And the conv_2d with small kernel is adopted to further extract multi-scale temporal features.

C. Spatial Convolution Block

The different EEG channels collect signals by electrodes from different brain regions, which contains wealth of spatial information. The spatial convolution layer with kernel size equivalent to channel number is served as the spatial filter to extract spatial features of EEG, and the batch normalization is used to enhance the stability. Then the maxpooling layer is applied to reduce the feature map dimension. Finally, these features are flattened and fed into the fully convolution layer with linear classifier.

III. EXPERIMENT

A. Dataset

The BCI competition IV 2a dataset [11] is used in the experiment to evaluate the effectiveness of the proposed method. This dataset collects EEG signal from nine subjects, which conduct four types of MI tasks, including left hand, right hand, feet and tongue. The signal is recoded in two sessions with 288 trials, which are treated as train (T) and test (E) data respectively. And the EEG signal contains 22 channels, with the sampling rate 250 Hz. In the EEG signal preprocessing, the Butterworth bandpass filter with bandwidth [4,40] Hz is used to filter the original EEG signal to improve the SNR and reduce out of band interference.

B. Implementation details

We implement the proposed MODWT-Net based on the Pytorch framework. The configuration of the model is shown in Table 1. And we can see that the total number of the MODWT-net network parameter is 4360, which is much smaller than many current DL-based decoding methods.

TABLE I. THE NETWORK PARAMETER OF THE MEHOD

Layer	Kernel size	Param
Attention	-	10
Conv2d	(1, 1)	450
BatchNorm2d	-	60
Conv2d	(1, 35)	1050
BatchNorm2d	-	60
Conv2d	(1, 25)	750
BatchNorm2d	-	60
Dropout	-	0
Conv2d	(1, 1)	300
BatchNorm2d	-	20
Conv2d	(1, 10)	100
BatchNorm2d	-	20
Conv2d	(C,1)	220
BatchNorm2d	-	20
Maxpooling	(1,25)	0
Linear	(1, T//10)	1240

In the training process, the model uses the cross-entropy as the loss function, which defines the differences between prediction and true labels. The Adam optimizer with the learning rate 0.001 and epoch 300 is used to update network weights in the data training. To compare the performance of different models, we use the classification accuracy (Acc) and kappa value as the evaluation metrics.

C. Results

We compare the proposed method with other network models, including FBCSP [4], Deep ConvNet [5], Shallow ConvNet [5], EEGNet [8], FBCNet [9] and Conformer [12], and the experimental result on BCI competition IV 2a is shown in Table 2. The proposed MODWT-Net achieves the best precision with the average accuracy 78.7% and kappa 0.7, and obtains balanced performance on the nine individual subjects, which have shown that our method outperforms other networks significantly. Therefore, the MODWT-Net has been demonstrated effective in EEG signal classification with light-weight network.

TABLE II. COMPARITION RESULTS OF DIFFERENT METHODS

Method	Subject										
	A01	***A02***	***A03***	***A04***	***A05***	***A06***	***A07***	***A08***	***A09***	***Acc(%)***	***kappa***
FBCSP	76.7	58.6	81.2	57.6	38.5	48.2	76.3	79.1	78.8	66.1	0.54
Deep ConvNet	76.4	55.2	89.2	74.6	56.9	54.1	92.7	77.1	76.4	72.5	0.63
Shallow ConvNet	74.7	58.0	85.2	67.7	68.4	54.9	83.0	86.5	79.5	73.1	0.64
EEGNet	87.5	65.2	90.2	66.6	62.5	45.4	59.5	88.3	79.5	74.4	0.65
FBCNet	85.4	60.4	90.6	76.4	74.3	53.8	84.4	79.5	80.9	76.2	0.68
Conformer	88.1	61.4	93.4	78.2	74.3	52.0	65.2	92.3	88.1	78.6	0.71

MODWT-Net	86.8	64.2	90.2	74.0	67.1	62.5	89.9	85.8	87.8	78.7	0.71

IV. CONCLUSION

In this work, we propose a MODWT and light-weight CNN combined EEG decoding method. This model integrates time-frequency and temporal-spatial features effectively, thus fully extracting and exploiting the most subject-related features of MI-EEG signal to improve the performance. The proposed method is validated on the BCI competition IV 2a dataset, the comparison results with state-of-the-art methods has proven the MODWT is effectively and robust in EEG signal decoding and classification.

REFERENCES

[1] J. W. Choi, B. H. Kim, S. Huh, and S. Jo, “Observing actions through immersive virtual reality enhances motor imagery training,” IEEE Trans. Neural Syst. Rehabil. Eng., vol. 28, no. 7, pp. 1614–1622, Jul. 2020.

[2] U. Chaudhary, B.S. Chander, A. Ohry, A. Jaramillo-Gonzalez, D. Lulé, N.Birbaumer, “Brain computer interfaces for assisted communication in paralysisand quality of life,” Int. J. Neural Syst. 31 (11), 2130003, 2021.

[3] P. Stegman, C.S. Crawford, M. Andujar, A. Nijholt, J.E. Gilbert, “Brain–computer interface software: a review and discussion,” IEEE Trans. Hum.-Mach. Syst. 50 (2) , pp.101–115, 2020.

[4] K. K. Ang, Z. Y. Chin, C. Wang, C. Guan, and H. Zhang, “Filter bank common spatial pattern algorithm on BCI competition IV datasets 2a and 2b,” Frontiers Neurosci., vol. 6, p. 39, Mar. 2012.

[5] R. T. Schirrmeister et al., “Deep learning with convolutional neural networks for EEG decoding and visualization,” Hum. Brain Mapping, vol. 38, no. 11, pp. 5391–5420, 2017.

[6] Qin, Yuxin, et al. "ETCNet: An EEG-based motor imagery classification model combining efficient channel attention and temporal convolutional network." Brain Research 1823 (2024): 148673.

[7] CAI, Zikun, et al. "Multi-branch spatial-temporal-spectral convolutional neural networks for multi-task motor imagery EEG classification." Biomedical Signal Processing and Control, 2024, 93:106156.

[8] V. J. Lawhern, A. J. Solon, N. R. Waytowich, et al. “EEGNet: A compact convolutional neural network for EEG-based brain–computer interfaces,” J. Neural Eng., vol. 15, no. 5, Oct. 2018, Art. no. 056013.

[9] R. Mane, N. Robinson, A. P. Vinod, et al. “A multiview CNN with novel variance layer for motor imagery brain computer interface,” in Proc. 42nd Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC), Jul. 2020, pp. 2950–2953.

[10] Tajmirriahi, Mahnoosh, et al. "An interpretable convolutional neural network for P300 detection: Analysis of time frequency features for limited data." IEEE Sensors Journal 22.9 (2022): 8685-8692.

[11] M. Tangermann et al., “Review of the BCI competition IV,” Frontiers Neurosci., vol. 6, p. 55, 2012.

[12] Y. Song, Q. Zheng, B. Liu, and X. Gao, “EEG conformer: Convolutional transformer for EEG decoding and visualization,” IEEE Trans. Neural Syst. Rehabil. Eng., vol. 31, pp. 710–719, 2023.